Field inspection manual—volumetric measuring devices

Part 3, section 8— Product depletion test

Application

This test is only necessary for meters that normally drain a tank completely, such as vehicle-mounted meters (pump or gravity), milk receiving systems, milk pick-up systems and other receiving / delivering systems.

Purpose

A product depletion test verifies the proper operation of air elimination means when the storage tank or supply lines for the product being measured is pumped dry.

Legislative references

R.262, R.265 to R.267, NOA, ruling established by the Volumetric Standing Committee (Administrative Limit of Error – ALOE).

Procedure for a multi-compartment tank (delivery systems e.g. truck-mounted meters)

1. At the normal operating rate of the meter, start the product depletion test from a compartment containing less test liquid than the capacity of the prover.

Note: In the case of metering systems with more than one outlet downstream of the meter, this test must be conducted on the outlet which offers the potential for the highest flow rate. For truck-mounted metering systems with a hose reel and a quick pump off, the quick pump off is considered the correct outlet to conduct this test through.

Note: In the case of metering systems with multiple product compartments, the compartment furthest from the pump, as measured along the piping, shall be used at the time of initial inspection. Subsequent inspections may be conducted using any other compartments as convenient, but the inspector should endeavour to use the compartment with the longest piping whenever practicable.

2. Continue the pumping until lack of liquid supply causes the register to stop indicating or until a maximum of 30 seconds has elapsed with no product flow, whichever comes first.

   Note: Running the pump dry for an extended period of time may damage it.

3. Without shutting off the pump, open the internal or emergency valve followed by the manifold valve from a compartment that contains sufficient liquid to complete the test. Leave the valves of the empty compartment open as this is a way of operating that could allow air to be introduced into the system. Some semi-automated systems will not allow multiple product paths (internal and manifold valves) to be open at one time. If this is the case, proceed with the test leaving only the appropriate valves open. Do not forget to close all valves after the test is complete. The intent is to test the system in a worst case configuration.

4. Continue the delivery until the liquid level is in the readable portion of the prover neck.

5. Compare the meter registration reading to the volume actually delivered into the prover. The difference (minus any meter error previously identified at the same rate of flow) is the error in the system under compartment switching conditions.

   Note: The product depletion test is the same for gravity discharge meters except there is no pump in the system.

Procedure for a single compartment tank (delivery systems e.g. truck-mounted meters)

Note: In the case of vehicle-mounted meters using a single compartment tank, this test is mandatory at the time the initial inspection is performed. At subsequent inspections, this test is performed when possible or when the inspector suspects that the air eliminator is defective. This test can be performed where there is a quick-connect hose coupling upstream of the meter. If there is no quick-connect hose, begin the procedure with a test quantity in the compartment less than the capacity of the prover. Pump the liquid until lack of liquid supply causes the register to stop indicating or until a maximum of 30 seconds has elapsed, whichever comes first. Then, complete the test by adding to the compartment the necessary quantity of liquid. This alternative test method requires a second tanker truck or the test is performed at a loading rack.

Note: In the case of metering systems with more than one outlet downstream of the meter, this test must be conducted on the outlet which offers the potential for the highest flow rate. For truck-mounted metering systems with a hose reel and a quick pump off, the quick pump off is considered the correct outlet through which to conduct this test.

Make a partial delivery from a flooded, primed system. During the test run (when the prover is approximately half full), close the outlet valve from the tank.

Break the hose connection, if possible, and let the pump drain the line. Continue the test until lack of liquid supply causes the register to stop indicating or until a maximum of 30 seconds has elapsed, whichever comes first.

Reconnect the supply line, open the valve and complete the test run by filling the prover.

Compare the meter registration to the volume actually delivered into the prover. The difference (minus any meter error previously identified at the same rate of flow) is the error that may be caused by air entrained into the system, poor re-priming conditions, faulty or plugged air eliminator, or other factors.

Procedure for metering systems using the same pump and lines to load and unload products

Note: This type of system is typically found at loading facilities or airports.

Note: This test is performed following the standard performance tests (fast test, slow test, etc).

1. Ensure that the installation complies with the requirements (selector valves, air eliminator) such that the system cannot load products through the meter while unloading product to the feed tank.
2. Set the selector valves to unload product to the feed tank. Turn the pump on and unload a small quantity of product. The product can be unloaded from a tanker truck or the test prover if this is convenient. Let the pump run until the suction line is empty. Close the valves.

3. Set the selector valves to the loading mode.
4. Run a fast flow test without re-priming the system.
5. Record the result and compare it to the result of the standard fast flow test. The difference (minus any meter error previously identified at the same rate of flow) is the error due to the product depletion test. These errors may be caused by air entrained into the system, poor re-priming conditions, inadequate air elimination system, or other factors.

Procedure for milk receiving and milk pick-up metering systems

Note: These systems must be capable of maintaining a constant transfer point.

Certain systems, called Pressure Air Elimination Systems or Type A Systems, are designed with their suction line connected to the inlet of a centrifugal pump; the outlet of the pump is connected to the inlet of an air eliminator and then to the meter. The air eliminator is maintained under positive pressure during the measurement operation. The transfer point is at the pump, so that the conduit between the pump and the meter remains full at all time (during the entire measurement process).

In the case of other systems called Vacuum Air Elimination Systems or Type E Systems, the air eliminator comes first and is located to have its inlet at a lower elevation than the product source. These air eliminators are equipped with a liquid level sensor that slows/shuts the pump and the flow valve off when the flow of milk slows or is exhausted. The transfer point is located in the air eliminator. The liquid flow is stopped when the liquid level in the air eliminator reaches the low stop position. The pump is located downstream from the air eliminator so that the air eliminator operates under vacuum at full flow or under low pressure. The meter is located downstream from the pump.

The entire configuration of milk receiving and milk pick-up systems, including the pump, the air eliminator, the meter, the valves and other associated equipment, must be approved as a system. Therefore, these systems must be installed as per the layout found in the Notice of Approval.

The air elimination system must prevent the passage of air through the meter. Air ingress becomes more critical when the operator switches from a drained compartment to a full compartment.

Since the metering system is used to empty various trucks with a number of different piping configurations, the relative drainage capability of various trucks is also evaluated. For example, a 2-inch side loading truck unloading into a 3-inch metering system in a sloped receiving bay would be more likely to fail the test than a 3-inch rear loading tanker unloading into the same system.

Note: The following procedure uses a 1500 litre sanitary narrow neck prover. It is recognized that sanitary provers with larger capacities are also used by industry.

1. Request a truck with one empty compartment that best represents the average tanker most likely to unload through the metering system.
2. Perform fast flow tests and determine the meter error in the usual manner. During the last test, deliver a volume approximately equal to half the prover capacity into an empty compartment.
3. Start the product depletion test into the prover from the now partially filled compartment from step 2 and continue pumping until the register stops advancing or for 30 seconds after the compartment runs dry, whichever is less.
4. Open the valve of a full compartment, close the valve of the empty compartment, and finish the test.

5. Compare the meter registration to the volume actually delivered into the prover. The difference (minus any meter error previously identified at the same rate of flow) is the system error from the out of product test. These errors may be caused by air entrained into the system, poor re-priming conditions, inadequate air elimination systems, or other factors.

Note: It is not always possible to obtain a truck with an empty compartment. As an alternative, a product depletion test can be simulated by closing the compartment valve from which milk is being drawn and loosening the hose fitting on the manifold (opening the hose to atmosphere) until the system has automatically stopped indicating (Type E), or for 30 seconds (Type A). The fitting must then be re-tightened and the valve re-opened to finish the test.

Procedure for unloading (receiving or pick-up systems) metering systems other than milk systems

Note: This type of system is used to measure products at the time of unloading trucks or railway tank cars.

Note: Unlike milk receiving systems, these systems are not type approved. They are custom designed. There are two critical points that can impact accuracy:

- the efficiency of the de-aeration / air elimination system,
- the use of a consistent transfer point.

The design and composition of these systems must be closely examined; tests must be performed to make sure that, under normal operating conditions, they measure accurately.

Systems can be designed with their suction line connected to the inlet of a pump; the outlet of the pump is then connected to the inlet of an air eliminator, the outlet of which is connected to the inlet of the meter. The air eliminator is maintained under positive pressure during the measurement process. The transfer point is at the valve immediately adjacent to the pump, so that the piping between the pump and the meter remains full at all time (during the whole measurement process). The system must be provided with automatic means to shut the pump off and close the valve immediately when the liquid is exhausted in order to maintain a constant transfer point.

On other systems, the air eliminator comes first and is located with its inlet at a lower elevation than the product source. These air eliminators must be equipped with a liquid level sensor that stops the pump and closes the flow valve when the flow of product slows or is exhausted. The transfer point is located in the air eliminator. The liquid flow is stopped when the liquid level in the air eliminator reaches the low stop position. The pump is located downstream from the air eliminator such that the air eliminator operates under vacuum or a very low pressure from the product head pressure. The meter is located downstream from the pump.

Lack of a consistent transfer point may affect accuracy significantly.

1. Request a vehicle (truck, railway tank car) with one empty compartment that best represents the average tanker most likely to unload through the metering system.
2. Ensure that the system is primed before proceeding with the tests.
3. Perform a fast flow test to establish the meter error. Record the results. Empty and drain the prover.

4. Perform another fast flow test. When the prover is partly filled, close the outlet valve from the source tank. Open an empty compartment which is connected to a common manifold with the source tank (or break the hose connection, if possible), and let the pump drain the line. Continue the delivery until a lack of liquid supply causes the register to stop indicating or until a maximum of 30 seconds has elapsed, whichever comes first.
5. Switch back to the full compartment (or reconnect the supply line), open the outlet valve from the source tank and complete the test run by filling the prover.
6. Compare the meter registration to the volume actually delivered into the prover. The difference (minus any meter error previously identified at the same rate of flow) is the system error from the product depletion test. These errors may be caused by air entrained into the system, poor re-priming conditions, inadequate air elimination systems, or other factors.

Interpretation of results

Note: The results of these tests may be significantly influenced by the repeatability of the meter itself. This should be taken into consideration in the assessment of the results.

The following administrative limits of error (ALOE) apply to any vehicle-mounted systems and to any receiving systems other than milk receiving or milk pick-up systems.

The administrative limit of error between a fast test and a product depletion test is the absolute value of the limit of error otherwise applicable to the meter under test at the following stipulated volumes:

- Where the meter is a 65 mm (2.5 inches) or smaller meter, **the ALOE is based on a test volume of 900 litres**.
- Where the meter is a 75 mm (3 inches) or larger meter, **the ALOE is based on a test volume of 1500 litres**.

For example, under Regulation 266, the in-service limit of error for a meter is 0.25%. Therefore, the allowable administrative limit of error for product depletion is:

- Where the meter is a 65 mm (2.5 inches) or smaller meter, **the ALOE is the base test volume of 900 litres × 0.25% = 2.25 litres**.
- Where the meter is a 75 mm (3 inches) or larger meter, **the ALOE is the base test volume of 1500 litres × 0.25% = 3.75 litres**.

The administrative limits of error would be changed accordingly for meters subject to other tolerances.

For example, under Regulation 267, the in service limit of error is 0.50%. Therefore, the allowable administrative limit of error for product depletion is:

- Where the meter is a 65 mm (2.5 inches) or smaller meter, **the ALOE is the base test volume of 900 litres × 0.50% = 4.50 litres**.
- Where the meter is a 75 mm (3 inches) or larger meter, **the ALOE is the base test volume of 1500 litres × 0.50% = 7.50 litres**.

The following administrative limits of error for the product depletion test apply to milk receiving and milk pick-up systems.

The difference between the normal fast flow test and the product depletion test must not exceed 0.25% of the known test volume. All test results, including product depletion test results, must be within the in-service limits of error prescribed by section 266 of the Weights and Measures Regulations.

Revisions

Revision 3

- Clarify compartment switching procedure for bulk fuel trucks including the addition of the requirement to use the furthest compartment from the pump.
- Clarify the administrative limits of error (ALOE) and associated examples.
- Reword certain sections including Interpretation of Results to make the intent clear.
- Reformat bulletin to reflect new standard practices.

Revision 2

- The Interpretation of Results and explanation of the administrative limit of error were revised to clarify the application to other products and applicability of other limits of errors. The name of the test procedure was changed from "Split Compartment / Out of Product Test" to "Product Depletion Test" to standardize and harmonize the terminology.

Revision 1

- Notes under "Procedure for a Multi-compartment Tank (Delivering Systems Such as Truck-Mounted Meters)" and "Procedure for a Single Compartment Tank (Delivering Systems Such as Truck-Mounted Meters)" were added to clarify through which outlet this test should be performed on for systems with more than one outlet.

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